Assignment 1: Self-Supervised Learning with SimSiam on STL-10

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Abstract

We re-implement SimSiam [2] with a ResNet-18 backbone and pretrain it on the 100k unlabeled images of STL-10. With only 1,000 labeled images, a linear probe on the frozen SimSiam encoder reaches $80.98 \pm 0.35\%$ test accuracy (200 epochs, 3 pretraining seeds), while the same network trained from scratch reaches only $47.74 \pm 1.17\%$. We also reproduce the complete collapse reported in the original paper: removing either the stop-gradient or the predictor drops the linear probe to exactly chance level (10.00%). As a new ablation, we remove one family of augmentations at a time. Without cropping, SimSiam does not collapse to a point but suffers dimensional collapse: the backbone's effective rank falls from 318 to 6.7, and accuracy drops with longer training. Replacing the crop with simpler operations shows that the key is the difference between the two views: sharing one random crop between both views is as bad as no cropping, while independent random masking recovers two thirds of the gap.

1. Introduction

SimSiam [2] is one of the simplest self-supervised methods. It trains a Siamese network to make the representations of two augmented views of an image agree, and uses no negative pairs, no large batches and no momentum encoder. Contrastive methods such as SimCLR [1] rely on negatives to avoid the constant solution, BYOL [5] on a momentum encoder; SimSiam claims that a stop-gradient and a predictor alone suffice.

Without negatives, the difference between the two augmented views is SimSiam's only learning signal. SimCLR found crop plus color the most important augmentation pair for contrastive learning, and InfoMin [12] argues that good views should share task-relevant information and differ in everything else. The SimSiam paper, however, ablates the architecture and the loss but not the augmentations that create this signal. In this report we re-implement SimSiam on STL-10 [3] and study both sides. We (i) show that SimSiam pretraining on 100k unlabeled images beats training from scratch by 33.2 points with 1,000 labels (Sec. 5.2); (ii) reproduce the complete collapse caused by removing the stop-gradient or the predictor (Sec. 5.3); (iii) show that without cropping SimSiam suffers dimensional collapse that worsens with training and that neither the loss nor the output std detects (Sec. 5.4); and (iv) decompose the crop and find that what matters is a content difference between the two views, not data diversity (Sec. 5.5).

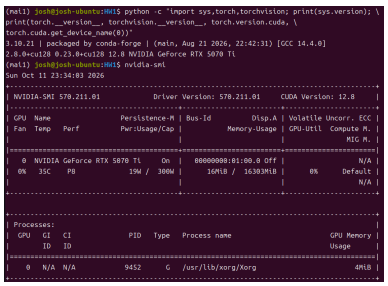


Figure 1. Experimental environment (nvidia-smi).

2. Environment and Codebase

All experiments ran on one NVIDIA GeForce RTX 5070 Ti (16 GB, driver 570.211.01, CUDA 12.8) with Python 3.10.21, PyTorch 2.8.0 and torchvision 0.23.0 (Fig. 1). The code extends the course starter code:

- model.py: SimSiam model and loss, with stop_grad and predictor flags for the ablations.
- data.py: STL-10 loaders, validation split and the two-view augmentation with an aug option (Sec. 3).
- pretrain.py: pretraining; logs loss, kNN accuracy and output std every epoch.
- linear_probe.py: linear (or BN) probe on the frozen backbone.
- sft.py: the same ResNet-18 trained from scratch; utils.py: LR schedule, kNN monitor, seeding, logging.
- Analysis: plot.py/visualize.py (figures), analyze_rank.py (erank), eval_probes.py (AUROC/F1), stats_tests.py (t-tests); run_*.sh drive all experiments.

3. Method

3.1. SimSiam

Given an image x, two random augmentations produce views x_1 and x_2. An encoder f, consisting of a backbone and a projection MLP, maps each view to $z_i = f(x_i)$, and a prediction MLP h maps it to $p_i = h(z_i)$. With the negative cosine similarity $\mathcal{D}(p, z) = -\frac{p}{\|p\|_2} \cdot \frac{z}{\|z\|_2}$, the symmetric loss is

$$\mathcal{L} = \tfrac{1}{2}\mathcal{D}(p_1, \mathrm{sg}(z_2)) + \tfrac{1}{2}\mathcal{D}(p_2, \mathrm{sg}(z_1)), \qquad (1)$$

where $\mathrm{sg}(\cdot)$ is the stop-gradient. The minimum of $\mathcal{L}$ is -1.

We use a ResNet-18 backbone [6] with its fc layer removed (512-d output), a 3-layer projection MLP ($512 \to 2048 \to 2048 \to 2048$, BN on every layer) and a bottleneck prediction MLP ($2048 \to 512 \to 2048$).

3.2. Ablations

Architecture. Following Sec. 4.1–4.2 of the original paper, we (a) remove the stop-gradient so that gradients flow through both branches, and (b) replace the predictor h with the identity.

Augmentation. The original paper's EM-like hypothesis (Sec. 5) views SimSiam as approximately solving

$$\min_{\theta} \; \mathbb{E}_{x,\mathcal{T}} \big\| f_\theta(\mathcal{T}(x)) - \eta_x \big\|^2, \qquad (2)$$

where f_θ is the encoder and η_x is the expected representation of x over augmentations $\mathcal{T}$. If the augmentations are too weak, this expectation becomes trivial. To test this, we remove one family of augmentations at a time: no_color drops color jitter and grayscale, and no_crop drops RandomResizedCrop, so that both views show the full image and differ only in flip and color.

Decomposing the crop. A crop shows only part of the content, rescales and shifts it, makes the two views differ and adds diversity across epochs. To separate these factors we replace it (keeping flip and color) with: crop_in_place keeps a RandomResizedCrop-style region at its original position and scale and fills the rest with the dataset mean (partial content only); zoom_out shrinks the whole image to area fraction a (crop-like aspect ratio) and pastes it at a random position on a mean-filled canvas (rescaling and shifting only); mask keeps a random fraction a of the 36 16×16 patches and fills the rest with the mean (partial content, scattered, no rescaling); same_crop applies one random crop to both views, so the data diversity matches the baseline but the views no longer differ in crop. In the first three, $a \sim U(0.2, 1)$ as in the crop, drawn independently for each view.

3.3. Measuring Collapse

Complete collapse maps every input to one vector; in dimensional collapse [9, 10] the embeddings stay spread out but occupy a low-dimensional subspace. The std of the ℓ_2-normalized output ($1/\sqrt{d}$ when spread over the sphere, 0 under complete collapse) detects only the former, so we also measure the effective rank (erank) [11], which RankMe [4] showed predicts downstream performance without labels. With σ_k the singular values of the centered, ℓ_2-normalized features of the 8,000 test images,

$$\mathrm{erank} = \exp\Big(-\sum_k p_k \log p_k\Big), \quad p_k = \frac{\sigma_k}{\sum_j \sigma_j}. \qquad (3)$$

We focus on the 512-d backbone feature, the probe's input.

4. Experimental Setup

Dataset. We use STL-10 [3] at 96×96. Pretraining uses the 100k unlabeled images. Classifiers train on the official fold 0 (1,000 images). Validation uses 500 held-out, label-balanced training images, and testing uses all 8,000 test images.

Pretraining. Following the original paper's CIFAR setting, we use SGD (momentum 0.9, weight decay 5×10^{-4}), batch size 512, lr $= 0.03 \times 512/256$, a 10-epoch warmup and cosine decay. The augmentation is RandomResizedCrop (scale 0.2–1), horizontal flip, color jitter $(0.4, 0.4, 0.4, 0.1)$ with p=0.8, and grayscale with p=0.2, without the Gaussian blur of its ImageNet setting. As in the paper, the predictor's lr is not decayed. A kNN monitor [14] (k=200, τ=0.1) uses fold 0 as the memory bank and evaluates on the validation images after every epoch.

Evaluation. The linear probe trains a linear layer on the frozen backbone with SGD (lr 30 as in MoCo [7], no weight decay) for 50 epochs. The BN probe adds a parameter-free BN before the linear layer as in MAE [8], with lr chosen on validation. The from-scratch baseline (SFT) trains the whole ResNet-18 with the pretraining optimizer settings (batch 512, about 2 steps per epoch) for 50 epochs. Both train with RandomResizedCrop (scale 0.08–1) and horizontal flip. The pretrained encoder is always the final-epoch checkpoint; for the classifier we test the probe/SFT epoch with the best validation accuracy.

Method	Pretrain	Test acc. (%)	n
SFT from scratch	–	47.74±1.17	5 ft. seeds
Linear probe	100 ep	75.80	1 run
Linear probe	200 ep	80.98±0.35	3 pt. seeds
Linear probe	1000 ep	86.74±0.23	1 run†

Table 1. Test accuracy with 1,000 labeled images. †One pretraining run, so ± covers finetuning seeds only.

Setting	Loss	Std	kNN	Lin. probe
Baseline	−0.908	0.021	77.6	81.35
w/o stop-grad	−1.000	0	10.0	10.00
w/o predictor	−1.000	0	13.2	10.00

Table 2. Architecture ablations after 200 epochs (final-epoch values; one unseeded run each, so the baseline differs slightly from Tab. 3). After 100 epochs, the two ablations reach 14.89% and 16.16% linear probe accuracy.

Statistics. For the 200-epoch comparisons we pretrain with seeds 123/456/789, and linear-probe each model with 5 finetuning seeds (12/34/56/78/90). The pretraining seed is the unit of replication: we report the mean ± std over the 3 per-model means and compare settings with two-sided Welch t-tests. We also compute macro one-vs-rest AUROC and macro F1 for every probe, and check every comparison under all three metrics with Holm correction. With one pretraining run, ± covers finetuning noise only.

5. Experiments

The original paper uses ImageNet and ResNet-50, so we compare only qualitative trends.

5.1. SimSiam Pretraining

Figure 2 shows a 100-epoch pretraining run. The loss falls to about −0.78 during the 10-epoch warmup and ends at −0.848; the kNN accuracy rises steadily to 73.4%. The std of the normalized projector output ends at 0.02195, close to $1/\sqrt{2048} = 0.02210$, so the model has not collapsed.

Between epochs 50 and 80 the loss stays flat and even rises slightly (−0.843 → −0.837), while the kNN accuracy climbs from 60.6% to about 70%. The loss only measures agreement between two views, not how useful the features are (see also Sec. 5.4).

5.2. Classification with and without Pretraining

Both classifiers use the same ResNet-18 and the same 1,000 labeled images (Fig. 3, Tab. 1). The linear probe passes 70% validation accuracy around epoch 10, while SFT stays near chance for 13 epochs and plateaus around 45%. After 200 epochs of pretraining, the linear probe beats SFT by 33.2 points (95% CI 31.8–34.7, Welch t=59.2, p=2.4×10^{-8}).

Training longer does not close the gap: SFT saturates at about 49–50% (100 epochs: 49.62%; 200 epochs: 49.01%, with training loss 0.24, so it overfits), and a 200-epoch linear probe adds only +0.8 points over 50 epochs. SFT's batch size and lr were not tuned, however (see Sec. 6). SimSiam therefore learns from the 100k unlabeled images what 1,000 labels alone cannot teach.

Error analysis. Errors are mostly between similar animals (cat↔dog, dog↔horse, deer↔horse; Fig. 4). Vehicles (car, ship, truck, airplane) reach 86–93% per-class accuracy. Context also misleads, e.g. an airplane against blue sky becomes "bird".

5.3. Ablation: Stop-Gradient and Predictor

Both variants reproduce the collapse reported in the original paper (Fig. 5, Tab. 2). Within a few epochs the loss reaches its minimum of −1 and the output std drops toward 0, so every image is mapped to the same vector. After 200 epochs the linear probe scores exactly 10.00% with training loss $\ln 10 = 2.303$, which is chance level for 10 classes. The original paper's 0.1% is the same result for ImageNet's 1000 classes. After 100 epochs the probes still reach about 15%, because the backbone has not fully collapsed yet.

The two ablations produce nearly the same curves, which matches the original paper's analysis. Without a predictor, the gradient of Eq. (1) has the same direction as that of $\mathcal{D}(z_1, z_2)$, scaled by $\frac{1}{2}$, so removing the predictor in effect also removes the stop-gradient. Neither the architecture nor BN prevents the trivial solution; the asymmetry from the stop-gradient and the predictor is required, in line with the dynamics analysis of DirectPred [13]. Collapse here is deterministic, so one run per variant suffices.

5.4. Ablation: Augmentation

Lower loss, worse representation. Both variants reach a lower loss than the baseline, because the views are easier to align, yet their accuracy is far lower (Tab. 3). With the same probe and n=3 on both sides, removing color costs 30.6 points (p=1.8×10^{-4}), removing crop costs 63.5 points (p=3.0×10^{-4}), and no_crop is 32.9 points below no_color (p=1.6×10^{-4}). The standard probe underestimates no_crop (below), but within seed 123 the BN probe gives the same ordering: baseline 82.4, no_color 51.2, no_crop 25.4. Cropping

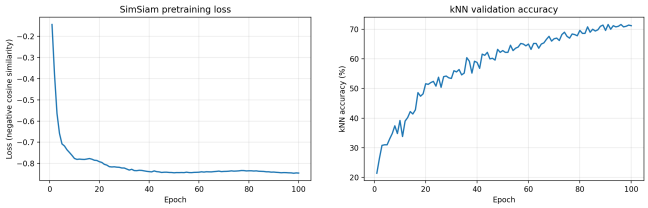


Figure 2. SimSiam pretraining for 100 epochs: training loss (left) and kNN validation accuracy (right).

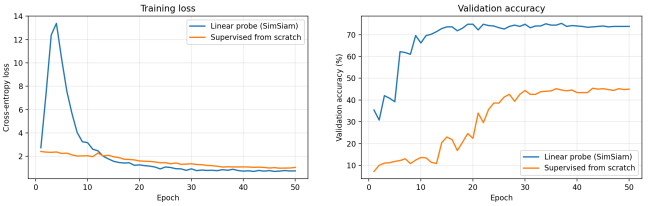


Figure 3. Linear probe on the frozen 100-epoch encoder vs. SFT on the 1,000 fold-0 images: training loss (left) and validation accuracy (right).

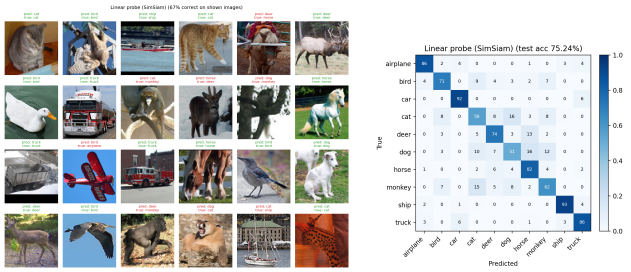


Figure 4. Linear probe on random test images (left; green: correct, red: wrong) and its confusion matrix on all 8,000 test images (right). Errors concentrate among animal classes.

Setting	Loss	kNN	Lin. probe	BN probe
Baseline	−0.909	78.1	80.98±0.35	82.36†
w/o color	−0.958	44.9	50.36±1.17	51.18†
w/o crop	−0.995	19.3	17.50±2.19	24.77±1.54

Table 3. Augmentation ablation, 200 epochs, mean (±std) over 3 pretraining seeds. Loss and kNN are final-epoch values averaged over seeds. †Seed 123 only (no_crop seed 123: 25.41), so the BN column is compared only within seed 123.

is therefore the more important augmentation. This agrees with SimCLR, but the drop is larger here: without negatives, a too-easy view pair leaves no other signal to learn from.

The standard probe fails on no_crop. With lr 30, the linear probe on no_crop features never gets below training loss $\ln 10$, even with 200 epochs: the features have tiny variance, so the fixed lr is poorly scaled. The BN probe standardizes the features (validation selects lr 1.0, versus 0.03 for the baseline) and raises no_crop to 24.77% (+7.27, paired t=19.2, p=0.003). We therefore use the BN probe for no_crop below; its features remain 57 points below the baseline's under the same probe (seed 123).

no_crop gets worse with more training. no_color improves from 100 to 200 epochs (43.9 → 50.4), but no_crop moves the other way. With 3 seeded 100-epoch runs and the BN probe, its accuracy drops from 38.17 ± 2.39 to 24.77 ± 1.54 (Welch t=8.18, df 3.4, p=0.0023; 95% CI of the change $[-18.3, -8.5]$). The 100- and 200-epoch models come from separate cosine schedules, but the decline also shows within each 200-epoch run: no_crop's kNN accuracy peaks at 27–33% around epochs 15–21 and ends at 15–21%, while the baseline's keeps rising (Fig. 6, left).

Dimensional collapse, not complete collapse. The effective rank explains these results (Fig. 6, right). The 200-epoch no_stop-grad and no_predictor models are single points: their feature variance relative to the mean squared norm is 10^{-15} or smaller. no_crop is different. Its projector output z still spreads out (relative variance 0.93, nonzero std), but occupies only about 4–8 of 2048 dimensions, and its backbone erank is just 6.7 ± 2.8, versus 317.5 ± 2.4 for the baseline. This is dimensional collapse [10]. It worsens with training: for every seed, the backbone erank drops from 100 to 200 epochs (53.3 → 9.2, 120.3 → 7.4, 34.2 → 3.6) together with the accuracy. In terms of Eq. (2), a crop-free pair differs only in color and flip. Averaging over such augmentations gives an η_x that a few color-invariant directions can match, and the objective keeps shrinking the representation onto those directions. Stop-gradient and the predictor prevent complete collapse, but not dimensional collapse. Jing et al. [10] found the opposite trigger in contrastive learning: there, strong augmentation, whose variance exceeds the data's, combined with the implicit regularization of deep networks collapses dimensions. Here a weak augmentation collapses them by a different route: η_x becomes so easy to match that a few directions suffice.

Not overfitting. no_crop's loss on 8,000 unseen images is within 0.001 of that on 8,000 training images, with the same erank (9.0 vs. 9.1). All t-tests above

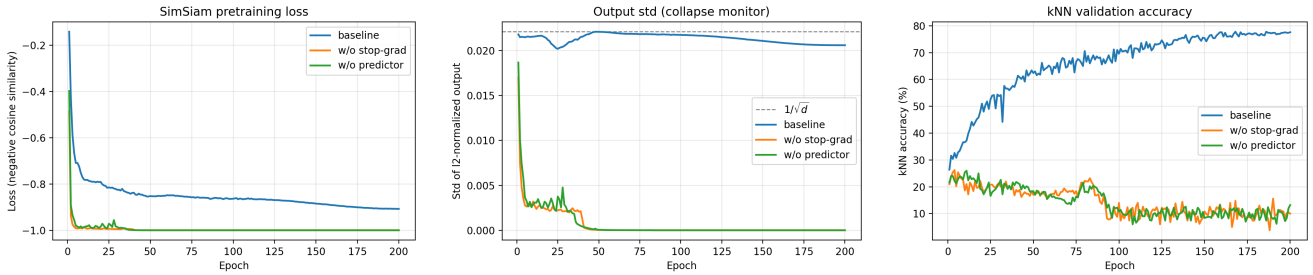


Figure 5. Ablating the stop-gradient and the predictor (200 epochs): training loss, std of the ℓ_2-normalized output (dashed: $1/\sqrt{d}$) and kNN accuracy. Both ablations collapse.

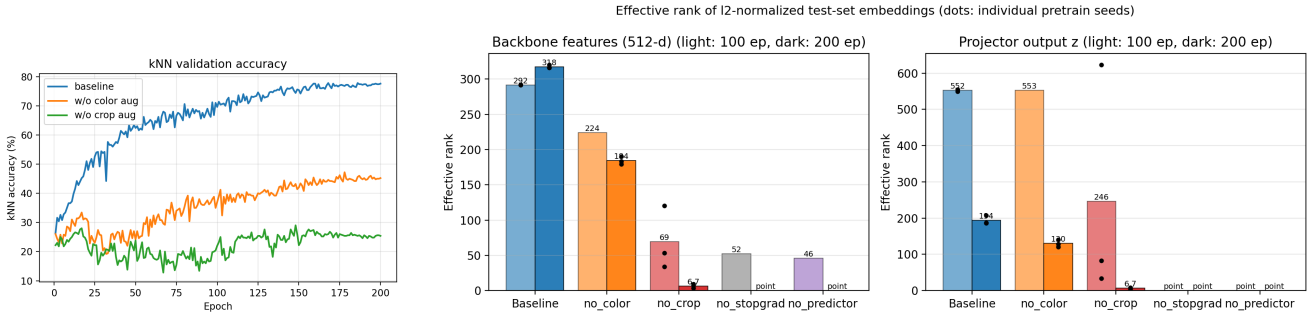


Figure 6. Left: kNN accuracy during augmentation-ablation pretraining (one representative 200-epoch run each). Right: effective rank of test-set embeddings for the backbone features and projector output z after 100 epochs (light) and 200 epochs (dark); bars are means and dots individual pretraining seeds. "point" means all images map to one point (relative variance $< 10^{-4}$), so no rank is defined; the 100-epoch no_stopgrad/no_predictor backbones are not yet points (relative variance 0.015–0.047).

Crop replacement	Loss	kNN	BN probe	AUROC
Full crop (baseline)	−0.909	77.8	82.36±0.64	98.00
mask	−0.953	53.2	63.41±0.64	93.95
crop_in_place	−0.947	38.6	49.81±0.88	88.71
zoom_out	−0.989	35.8	39.00±1.00	83.26
same_crop	−0.997	21.2	22.20±0.55	75.93
No crop	−0.995	21.2	25.41±1.38	72.25

Table 4. Replacing RandomResizedCrop (200 epochs, pretraining seed 123 only, so ± covers 5 finetuning seeds). Loss and kNN are final-epoch values; AUROC is macro one-vs-rest for the BN probe.

stay significant after Holm correction under accuracy, AUROC and F1 ($p_{\text{Holm}} \leq 0.035$).

5.5. What Does the Crop Contribute?

Table 4 reports the variants of Sec. 3, each pretrained once (seed 123) for 200 epochs. Measured as the fraction of the no_crop→baseline gap they recover under the BN probe, masking recovers 67%, crop_in_place 43% and zoom_out 24%.

The views must differ, not just the data. same_crop sees exactly as many distinct crops over training as the baseline, yet it is as poor as no_crop (22.2% vs. 25.4%) with the same near-minimal loss (−0.997) and the same kNN accuracy (21.2%). Data diversity and sample count therefore do not explain the no_crop failure. What matters is that the two views of one image differ in which content they show.

Partial content and rescaling are complementary. Each component alone recovers only part of the gap: hiding content (crop_in_place) helps more than rescaling and shifting (zoom_out), but neither comes close to the full crop, which combines both. Scattered masking is the strongest single replacement, probably because it hides different parts of the whole object in each view instead of one contiguous region. This matches the InfoMin view [12]: the views should share the object but differ in what else they contain.

5.6. Effect of Training Length

Longer schedules keep helping, with diminishing returns: 75.8% (100 epochs), 81.0% (200) and 86.7%

(1000), matching the trend in Tab. 4 of the original paper. In the 1000-epoch run, loss and kNN accuracy plateau at epochs 300–500 (LR above half its peak), then snapshot probe accuracy rises from 80% to 87% as the LR anneals, consistent with an LR-limited plateau (we ran no constant-LR control).

6. Limitations

(i) Erank is not a quality measure across settings: from 100 to 200 epochs the baseline's projector erank falls 552 → 194 and no_color's backbone erank 224 → 184 while accuracy rises. Our no_crop evidence rests on the backbone, where rank and accuracy fall together by large amounts. (ii) Nor is a falling std: the 1000-epoch baseline's std halves to 0.0100, yet its erank (392) and accuracy (86.7%) are our highest. (iii) With n=3 pretraining seeds the Welch df is only 2–4, though every effect is many standard deviations in size. (iv) The crop variants (Sec. 5.5) have one pretraining seed, although adjacent variants differ by 11–17 points versus a seed std of 0.35; their mean-filled regions, identical in both views, could also act as a shortcut. (v) SFT reuses the pretraining batch size (512, about 2 steps per epoch) and lr without tuning, and we did not fine-tune the SimSiam encoder end to end.

7. Conclusion

With 1,000 labels, SimSiam features beat training from scratch by 33 points. Removing the stop-gradient or the predictor causes complete collapse. When the views are too similar (no cropping), SimSiam keeps a nonzero std and a lower loss, yet its features collapse onto a few dimensions and degrade with training; the effective rank and a properly scaled probe detect this. The views must differ in the content they show: a crop shared by both views is as harmful as no crop at all.

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